

MIRA — Trajectory Turning-Point Investigator

Health changes gradually. MIRA finds the moments that changed its direction.

The problem

Longitudinal records contain years of labs, symptoms, treatments, visits and life events, but rarely explain when a health trajectory actually changed. Later outcomes can also create hindsight bias.

The innovation

MIRA treats health as a trajectory investigation, not a snapshot. It identifies meaningful turning points, investigates surrounding evidence, reconstructs what was knowable then, and checks whether later actions are followed by recovery or continued reversal.

Core method

State → Signal → Turning Point → Evidence → Action → Outcome → New State

Evidence is labeled Observed, Associated, Contradictory, Unknown, or Hypothetical. A hindsight firewall prevents later events from leaking into an earlier decision point.

Prototype workflow

Find the change → Investigate why → Ask what was knowable then → Act/review → Re-measure → Verify recovery

Validation plan

Synthetic benchmark with known states, turning points, reversals, recovery, resilience, contradictions and delayed evidence. Measure turning-point precision/recall, reversal detection, evidence agreement, hindsight separation and recovery verification.

Safety & scope

Research/education prototype only. Synthetic data; no diagnosis or treatment prescribing. Future clinical use would require privacy/security review, expert validation, prospective evaluation and regulatory assessment.